Bullet Farm

A transaction processor around powerful coding agents

Executive brief — Stage-1 architecture and component assurance — 25 August 2026

Decision summary

What it is.

- A software-change transaction processor *around* coding agents (Claude, Codex, Cursor, Antigravity, others). Agents reason and propose patches; they never hold repository, verification, or merge authority.
- Four repositories (Hub, Kernel, BulletGit, Portal) across five trust planes (control, execution, verification, delivery, audit), so no plane can use its own assertion as evidence.

What it is worth if it works.

- *Teams running many agents* stop losing work to seven recurring failures: two writers on one checkout, a stale process that still writes, tests that name a moved branch, a timed-out push retried into a duplicate, silence read as death or success, cleanup before preservation, and a dashboard that paints missing state green (Gas Town #4094/#3584/#4737/#4091, Gas City #1181/#5511/#1052 are public instances).
- *Reviewers and auditors* get, for every integrated change, exact evidence of who observed which bytes under which admitted gate; a rebase, gate, or policy change invalidates it instead of carrying stale green.
- *Operators* swap models and vendors without changing who may write; cost, refusal, and unresolved unknowns are recorded per change. The target metric, accepted surviving change per human intervention and per dollar, is a Stage-2 measurement, not a claim here.
- *Later:* bounded multi-agent evolution over evidence lineages (Designed; post-V1).

What is proven today. Snapshot 20260825-stage1-r3: Hub 3039878371b6, Kernel ca380bc4ffd4, BulletGit fb715b25c3b2, Portal 95108e37d643.

- Component-proved only: fast PASS, demo PASS, contract, registry, and required lanes PASS, 2 bounded TLA+ safety models PASS.
- Not proved: `TRANSACTION_PROOF` ABSENT; live provider, forge, and release proof absent; doctor BLOCKED by design; 26 blocked of 26 release gates (`check release`, exit 3).
- Security audit 58 (raw 58), 10 caps, 29 hard findings; release floor 90/0/0.

Biggest concerns. Ranked; the paper names the receipt that retires each.

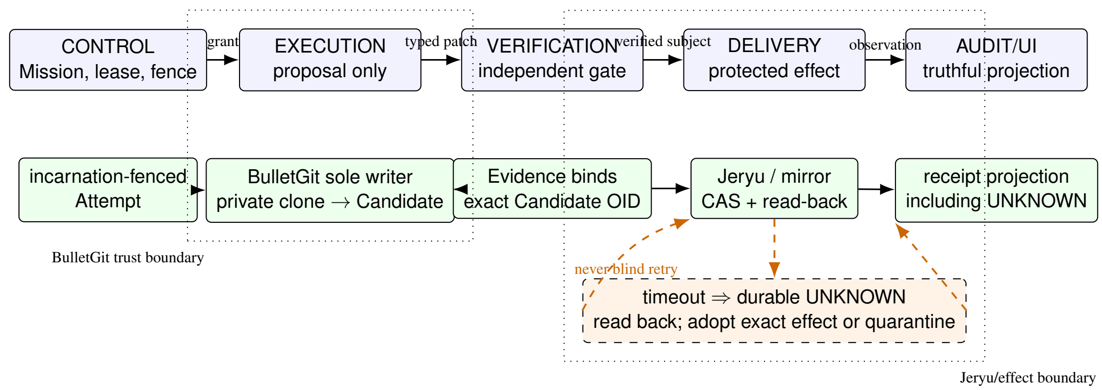
- Connected product unassembled: 0 of 26 gates receipted; every positive result is component-level; the family's own weighted self-assessment is 26/100 (a planning judgment, not a receipt).
- No live evidence at this subject: the committed policy refuses every provider before spawn; no authenticated Jeryu or GitHub read-back exists.
- Closure needs operator custody acts: live-admission ratification, Jeryu authentication, release signing keys, schema-3 lock inputs.
- Single host, Linux only, same UID; some Git/helper paths remain raceable; macOS/Windows refuse.
- Verifier independence rests on one fixture gate and human-curated holdouts; a later evolutionary layer inherits that dependence.
- Security score below its own floor; no portable hosted audit artifact.
- Split-brain, quota, and liveness are bounded, not eliminated; formal models are bounded safety only.
- Size, cost, adoption: five planes, 26 gates, private-clone storage, read-back latency; only Jeryu and a GitHub adapter specified; cost unmeasured.

What must happen before launch. Predecessor order; owner in parentheses.

- Operator ratifies live admission (policy generation 2, provider-runner key) and supplies Jeryu authentication, signing custody, schema-3 lock inputs (operator: G1, G5, G6, G7).
- One signed connected five-plane `TRANSACTION_PROOF` (engineering: G2–G4), then sealed live receipts for four providers and Jeryu/GitHub read-back (both: G5–G7).
- Audit at floor with a portable CI artifact; five signed packages with SBOM, provenance, checksums; installer twice in a fresh home; platform refusal receipts (engineering: G8–G10).
- Only then the Stage-2 matched benchmark, before any "faster, cheaper, safer" claim.

Why this is powerful: the binding chain

Four questions decide whether agent work survives: *who owns the write, which incarnation is current, which exact bytes were proved, and did the remote mutation really happen?* A sandbox, worktree, reviewer, or merge queue answers at most one, which is why an agent can reason brilliantly and still lose work when two sessions share a checkout, a retry duplicates a workflow, CI approves a branch that moved, or a dashboard reads silence as success. Bullet Farm puts all four answers outside the model; providers propose, and the repository remains sovereign even when models and vendors are swapped.



The design composes eight bindings: incarnation fence → sole writer → private Attempt clone → exact-OID Candidate → independent Evidence → durable effect intent → read-back/adoption → truthful projection.

1. **Concurrency becomes explicit.** A lease plus an incarnation fence means a process that wakes after reclaim is stale, even if it still runs.
2. **Git identity becomes exact.** One BulletGit writer applies typed patches in private Attempt clones and mints an immutable Candidate from exact base, head, tree, and provenance subjects.
3. **Evidence cannot float.** An independent verifier proves that Candidate, not a moving branch, a transcript, or a model's "done" statement.
4. **Timeout is survivable.** Durable effect intent plus read-back distinguishes applied, not applied, and honest `UNKNOWN`; an ambiguous push is never blindly retried.
5. **The UI cannot manufacture truth.** Projections retain stale, contradictory, failed, and unknown outcomes instead of painting them green.

A failure, recovered without guessing

Suppose a protected-ref update times out after the forge may have accepted it. A conventional retry loop assumes failure and sends the mutation again. Bullet Farm has already committed one logical effect identity, expected-old OID, intended-new OID, and Evidence root; it records `UNKNOWN` and reads back:

Observed state	Required action
Ref equals intended OID	Adopt the already-applied effect and seal the read-back receipt.
Ref equals expected-old OID	Retry the same logical effect identity under the current fence.
Ref equals neither	Quarantine divergence; preserve bytes and require adjudication.
No trustworthy observation	Remain `UNKNOWN`; do not infer success or failure.

The unit that matters is not a chat session but this recoverable journey from authority to exact Candidate, Evidence, protected effect, and observation.

Implemented now / not yet proved

Prominent limit. Every positive Bullet result below is component evidence. `TRANSACTION_PROOF`, live-provider/forge proof, and release proof are absent. The release inventory is BLOCKED: 26 blocked of 26 evaluated gates.

Boundary	Implemented now	Not yet proved
Control authority	DB-clock lease/fence, atomic command/outbox, quarantined signed lease-transport	Connected reservation/settlement from Kernel through Runner and BulletGit
Provider boundary	Offline protocol subsets, admission and egress components, policy-first refusal	Any live provider receipt; live policy remains disabled in the evidence subject
Repository writer	Fail-closed BulletGit gateway, private-clone/preservation components, strict manifests	Positive online sole-writer path; public clone still refuses as `AUTHORITY_CONTRACT_UNAVAILABLE`
Candidate/Evidence	Exact-subject fixture gate and bounded verifier transport	Production reconstruction, executable identity for all gates, multi-gate aggregation
Effects	Ambiguity model and local bare-forge components	Authenticated expected-old-OID push, read-back, adoption, and protected integration
Audit/Portal	Generated DTOs, exact command reconciliation, watermark projections, explicit `UNKNOWN`	Packaged complete product and every designed ledger subject
Formal assurance	2 bounded safety models, PASS at pinned state counts	Liveness, OS/Git/provider/forge composition, or a product proof
Release	Negative inventory; audit 58/raw 58, 10 caps, 29 hard findings	Quality floor, portable audit, packages, SBOM/provenance, signing, install/recovery, live receipts

How to read maturity

Designed	Reviewed rule; executable path may not exist.
Component-proved	Exact committed component plus mapped receipt; no cross-plane inference.
Transaction-proved	One exact connected offline five-plane transaction with independent receipts.
Live/release-proved	Admitted live provider/forge transaction, or signed package/install/recovery proof.

The snapshot reports doctor BLOCKED, fast PASS, demo PASS, and transaction proof ABSENT. Demo success is not silently relabeled as a transaction. A timeout, skip, unsupported path, missing receipt, or dirty-tree result cannot be promoted through prose.

The practical consequence

Bullet Farm is powerful as a control architecture now, but not yet proved as a connected product. That distinction is a feature: it makes investment and risk decisions against a visible boundary instead of a polished demonstration.

What adjacent systems contribute

Gas Town and Gas City provide strong orchestration, durable work, multiple runtimes, and merge-queue operations; DeepSeek Harness a composable plugin kernel plus an experimental Teams seam; Omnigent cross-vendor sessions, parallel worktrees and review, OS sandboxes, and an L7 credential proxy. Bullet Farm should retain those lessons. Its narrower claim is that the reviewed pinned public contracts do not document this same end-to-end authority/evidence/effect chain; linked worktrees, for example, share the object database and most refs and are not separate repository principals.

Publication, forge topology, and next decisions

Two publication stages

Stage 1 — now. Publish the architecture and component-assurance preprint, the four-page brief, bibliography, vector figure, JSON evidence snapshot, generated macros, and deterministic preflight. Make no comparative performance claim.

Stage 2 — later. Only after a signed connected `TRANSACTION_PROOF`, run a matched receipt-bearing corpus against pinned systems, recording tasks, subjects, models, budgets, retries, costs, collisions, duplicate effects, unknowns, recovery, and loss. Only this stage may support "faster," "cheaper," "safer," or "more successful."

Frozen V1 topology and blocked proposals

- V1 source distribution is Jeryu-only from immutable signed tags. The manifest admits no authenticated Jeryu URL and assumes no public mirror.
- The only permitted local Jeryu family is `/home/ubuntu/jain-split/jeryu-split`; its branded hub maps independent component repositories. Never recreate `/home/ubuntu/jeryu-split`.
- `git.neverhuman.org`, `github.com/neverhuman/jeryu`, and `github.com/neverhuman/jankurai` are unratified, operator-blocked future proposals. GitHub may later be chosen as a public index or backup, and independent Jeryu Git as a hosted front door; neither is authority today.
- Any later public mirror remains a separately configured effect adapter. It requires expected-old-OID CAS, read-back, divergence quarantine, and separately proved backups.
- Jankurai stays the existing monorepo at `/home/ubuntu/jankurai`. No public front door is assumed; preservation must precede any mirror decision.
- The local-family rule is frozen; the endpoint and mirror proposals are not. Operator ratification plus TLS, auth, protection, deployment, mirror, backup, and restoration receipts precede any name becoming authority.

Immediate decisions

1. Fund the connected five-plane transaction proof before comparative marketing.
2. Publish immutable signed `bullet-wire` only through an operator-ratified, authenticated Jeryu endpoint.
3. Select and prove protected Jeryu integration plus the ratified endpoint's operational controls.
4. Finish deterministic packages, SBOM/provenance, schema-3 installation, and portable Jankurai audit evidence.
5. Resolve Jeryu release-authority document drift and preserve Jankurai history before any mirror decision.
6. Publish immutable paper/source permalinks only after strict preflight is green from clean subjects.

Bottom line. More capable models increase the value of external authority, exact evidence, and recoverable effects. Bullet Farm has a defensible transaction architecture and meaningful components. It does not yet have the connected proof needed to claim a finished product or measured superiority.